Chapter 6

Eigenvalues and Eigenvectors

Homework list

Strang's Book

- Sec 6.1: 2, 3, 4, 6, 10, 12, 19
- Sec 6.2: 1, 2, 18, 27, 29 (Challenging for part (b): Prove that if AB = BA and A and B do not have any repeating eigenvalues, that they must be simultaneously diagonalizable.)
- Sec 6.4: 7, 25
- Sec 6.5: 36

Sec 6.1

Introduction to eigenvalues

What we have learned about $A\vec{x} = \vec{b}$:

- **Problems involving balancing production with demand, production with available resources, economic output, and chemical equations all gave rise to systems of equations that could be expressed as:**

$$A\vec{x} = \vec{b}$$

- **We can solve these systems using Gaussian elimination (LU decomposition) or finding approximate solutions with projections (QR decompositions).**
- **Both decompositions needs us to perform "expensive" calculation on A, then we solve for any right-hand sides with little effort:**

$$PA = LU \rightarrow \begin{matrix} L\vec{c} = P\vec{b} \\ U\vec{x} = \vec{c} \end{matrix}, \qquad A = QR \rightarrow R\vec{x} = Q^T\vec{b}$$

Key ideas:

- For a continuous time in a differential equation $\frac{dx(t)}{dt} = Ax$ or time steps in a differential equatio $A\vec{x}_i = \vec{x}_{i+1}$, we cannot solve by elimination.
  - **In Markov Chain model, we can find stationary values by setting $\vec{x}_i = \vec{x}_{i+1}$, which gives rise the homogenous system $(I - A)\vec{x} = \vec{0}$, but what about the more general case?**
  - **Can we avoid matrix multiplication and find a cheaper calculation to evaluate $\vec{x}_i$?** $A\vec{x} = k\vec{x} \rightarrow \vec{x}_{i+1} = A\vec{x}_i = k\vec{x}_i \rightarrow \vec{x}_{i+n} = A^n\vec{x}_i = k^n\vec{x}_i$
- **Key idea: We wanted to find a solution vector x(t) stays in the direction of a fixed vector x, i.e., a vector that don't change direction (scalar multiple) when you multiply by A.**

Eigenvalues and Eigenvectors

Definition:

- If a scalar λ and a non-zero vector $\vec{x}$ satisfy:

$$A\vec{x} = \lambda\vec{x}$$

- Then we say that λ is an **eigenvalue** of A, and $\vec{x}$ is an **associated eigenvector**.

Facts:

- When we multiply an eigenvector by A (on the left), the result is just a scalar multiple of the eigenvector.
- The scaling factor is the eigenvalue.
- Eigenvectors are always associated with some eigenvalue.

Applications of eigenvalues and eigenvectors

- **If we can express an arbitrary vector as a linear combination of eigenvectors, matrix multiplication becomes easy!**

$$\vec{u} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots, \qquad A\vec{x}_1 = \lambda_1\vec{x}_1, A\vec{x}_2 = \lambda_2\vec{x}_2, \cdots$$

$$\begin{aligned} A\vec{u} &= A(a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots) \\ &= a_1A\vec{x}_1 + a_2A\vec{x}_2 + \cdots \\ &= a_1\lambda_1\vec{x}_1 + a_2\lambda_2\vec{x}_2 + \cdots \end{aligned}$$

$$A^k\vec{u} = a_1(\lambda_1)^k\vec{x}_1 + a_2(\lambda_2)^k\vec{x}_2 + \cdots$$

- **Multiplication by any power of A is replaced by a sum of scalar multiplications.**

Applications of eigenvalues and eigenvectors

$$A^k\vec{u} = a_1(\lambda_1)^k\vec{x}_1 + a_2(\lambda_2)^k\vec{x}_2 + \cdots$$

- **When we multiply by large powers of A, only the largest (in absolute value) eigenvalue matters:**

$$|\lambda_1| > |\lambda_2| > |\lambda_3| \cdots \rightarrow \left|(\lambda_1)^k\right| \gg \left|(\lambda_2)^k\right| \gg \left|(\lambda_3)^k\right| \cdots, k \gg 1$$

$$A^k\vec{u} \approx a_1(\lambda_1)^k\vec{x}_1, k \gg 1$$

$$\lim_{k\to\infty} A^k\vec{u} = \lim_{k\to\infty} a_1(\lambda_1)^k\vec{x}_1$$

Questions?

1. **Can we always write a vector as a linear combination of eigenvectors?**
2. **Do all matrices have eigenvalues and eigenvectors?**
3. **If matrix has eigenvalues/eigenvectors, how many are there?**
   - **How do we find them?**

Finding eigenvalues.

- Is there a scalar λ and a non-zero vector $\vec{x}$ that satisfy $A\vec{x} = \lambda\vec{x}$?

$$A\vec{x} = \lambda\vec{x} \rightarrow \lambda\vec{x} - A\vec{x} = \vec{0} \rightarrow (\lambda I - A)\vec{x} = \vec{0}$$

- A must be square $\rightarrow$ only square matrices have eigenvalues.
- λ is an eigenvalue of A if and only if $(\lambda I - A)$ is singular:

$$\det(\lambda I - A) = 0$$

- $\det(\lambda I - A)$ is called the **characteristic polynomial** of A; its roots are the eigenvalues:
  - For an $n \times n$ matrix, it is a polynomial of degree n.
  - A polynomial of degree n can have at most n roots.
  - An $n \times n$ matrix can have at most n eigenvalues.

Ex 1: Finding eigenvalues.

- **Example: What are the eigenvalues of** $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$**?**

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 4$$

$$= \lambda^2 - 2\lambda + 1 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$$

$$\det(\lambda I - A) = 0 \rightarrow (\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

Ex 2: Finding eigenvalues.

- **Example: What are the eigenvalues of** $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$**?**

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 \\ +1 & \lambda - 2 \end{vmatrix} = \lambda(\lambda - 2) - (-1)$$

$$= \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2$$

$$\det(\lambda I - A) = 0 \rightarrow (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

Ex 3: Finding eigenvalues.

Quadratic formula:

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Example: What are the eigenvalues of** $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$**?**

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ +2 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - (-4)$$

$$= \lambda^2 - 2\lambda + 1 + 4 = \lambda^2 - 2\lambda + 5$$

$$\det(\lambda I - A) = 0 \rightarrow \lambda^2 - 2\lambda + 5 = 0 \rightarrow \lambda = \frac{2 \pm \sqrt{4 - 20}}{2}$$

$$\lambda = \frac{2 \pm \sqrt{-16}}{2} = 1 \pm 2i$$

The characteristic polynomial

For an $n \times n$ matrix, $\det(\lambda I - A)$ is a polynomial of degree n.

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & & \\ \vdots & & \ddots & \\ -a_{n1} & & & \lambda - a_{nn} \end{vmatrix}$$

- **Leading term is the product of diagonal entries:**

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn}) + \cdots$$

- **Highest order term is λ^n :**

$$\det(\lambda I - A) = \lambda^n + c_1 \lambda^{n-1} + \cdots + c_{n-1}\lambda + c_n$$

The characteristic polynomial

- **For an $n \times n$ matrix, $\det(\lambda I - A)$ is a polynomial of degree n.**
  - **A polynomial of degree n can have at most n roots.**
  - **Some roots may be repeated (degenerate).**
  - **Some roots may be imaginary or complex numbers.**
  - **Every polynomial has at least one root when we allow for complex numbers.**

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \rightarrow \det(\lambda I - A) = (\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \rightarrow \det(\lambda I - A) = (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

$$A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \rightarrow \det(\lambda I - A) = \lambda^2 - 2\lambda + 5 = 0 \rightarrow \lambda = 1 \pm 2i$$

- **An $n \times n$ matrix can have at most n distinct eigenvalues.**
- **Every $n \times n$ matrix has as least one eigenvalue, but they may be complex.**

Finding eigenvectors.

If λ is an eigenvalue, then $\det(\lambda I - A) = 0$ and $(\lambda I - A)$ is singular;

- There must be a non-zero vector $\vec{x}$ that satisfies $A\vec{x} = \lambda\vec{x}$.

$$A\vec{x} = \lambda\vec{x} \rightarrow (\lambda I - A)\vec{x} = \vec{0}$$

- This is a homogenous system; the solutions are the null space.
- Every vector in the null space of $(\lambda I - A)$ is an eigenvector of A with corresponding eigenvalue $\lambda \rightarrow$ an eigenspace.
- We characterize a vector space with a basis (eigenvectors) which is given by the special solutions to:

$$(\lambda I - A)\vec{x} = \vec{0}$$

- **Important:** Each distinct λ has a distinct eigenspace; we typically aim to find a basis (eigenvectors) for each eigenspace.

Finding eigenvectors

- **Example: Find a basis for the eigenspaces of** $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

$$\det(\lambda I - A) = (\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

$$\lambda = 3 \rightarrow (\lambda I - A) = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \rightarrow R = \vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda = -1 \rightarrow (\lambda I - A) = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow R = \vec{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$ **is a basis for the eigenspace associated with** $\lambda = 3$.
- $\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$ **is a basis for the eigenspace associated with** $\lambda = -1$.
- **These are linearly independent vectors in** $\mathbb{R}^2$**; together they span the whole space.**

Finding eigenvectors.

- **Example: Find a basis for the eigenspaces of** $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$.

$$\det(\lambda I - A) = (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

$$\lambda = 1 \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \rightarrow R = \vec{x}. = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$ **is a basis for the eigenspace associated with** $\lambda = 1$.
- **The eigenspace associated with the only distinct eigenvector has dimension 1 (it is a line); it does not span the whole space of** $\mathbb{R}^2$.

Finding eigenvectors.

- **Example: Find a basis for the eigenspaces of** $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$.

$$\det(\lambda I - A) = \lambda^2 - 2\lambda + 5 = 0 \rightarrow \lambda = 1 \pm 2i$$

$$\lambda = 1 + 2i \rightarrow \ (\lambda I - A) = \begin{bmatrix} 2i & -2 \\ 2 & 2i \end{bmatrix} \rightarrow \begin{bmatrix} 2i & -2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix} \ \rightarrow R = \vec{x} = \ \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

$$\lambda = 1 - 2i \rightarrow (\lambda I - A) = \begin{bmatrix} -2i & -2 \\ 2 & -2i \end{bmatrix} \rightarrow \begin{bmatrix} -2i & -2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix} \rightarrow R = \vec{x} \ = \ \begin{bmatrix} i \\ 1 \end{bmatrix}$$

- $\left\{\begin{bmatrix} -i \\ 1 \end{bmatrix}\right\}$ **is a basis for the eigenspace associated with** $\lambda = 1 + 2i$.
- $\left\{\begin{bmatrix} i \\ 1 \end{bmatrix}\right\}$ **is a basis for the eigenspace associated with** $\lambda = 1 - 2i$.
- **When we have complex eigenvalues, we get complex eigenvectors!**

Summary

- **Every square matrix has at least one eigenvalue and linearly independent eigenvector.**
- **An $n \times n$ matrix has at most n eigenvalues and at most n linearly independent eigenvectors.**
- **The solutions to $\det(\lambda I - A) = 0$ give us the eigenvalues of A.**
  - **This involves solving for the roots of a polynomial in λ.**
- **For each eigenvalue λ, the null space of $(\lambda I - A)$ is the eigenspace associated with λ.**
  - **We find a basis by finding the special solutions of $(\lambda I - A)\vec{x} = \vec{0}$.**

Examples

Finding the eigenvalues and eigenvectors of a matrix

Ex 1: Find the eigenvalues and eigenvectors.

- **Find the eigenvalues and a basis for each eigenspace for the matrix:**

$$A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

- **First find the roots of $|\lambda I - A|$(the eigenvalues):**

$$A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 & -3 \\ 0 & \lambda - 6 & 0 \\ -3 & 0 & \lambda - 3 \end{vmatrix}$$

$$= (\lambda - 3)\begin{vmatrix} \lambda - 6 & 0 \\ 0 & \lambda - 3 \end{vmatrix} - 0 + (-3)\begin{vmatrix} 0 & \lambda - 6 \\ -3 & 0 \end{vmatrix}$$

$$= (\lambda - 3)\big((\lambda - 3)(\lambda - 6) - 0\big) - 3\big(0 - (-3)(\lambda - 6)\big) = (\lambda - 3)^2(\lambda - 6) - 9(\lambda - 6)$$

$$(\lambda - 6)\big((\lambda - 3)^2 - 9\big) = (\lambda - 6)\big(\lambda^2 - 6\lambda + 9 - 9\big) = (\lambda - 6)\big(\lambda^2 - 6\lambda\big) = \lambda(\lambda - 6)^2$$

$$|\lambda I - A| = 0 \rightarrow \lambda(\lambda - 6)^2 = 0 \rightarrow \lambda = 0,6$$

Find the eigenvalues and eigenvectors.

- **Then find a basis for the null space of** $(\lambda I - A)$ **(the eigenspace) for each** λ**:**

$$\lambda = 6, A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 3 & 0 & -3 \\ 0 & 0 & 0 \\ -3 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & -3 \\ 0 & 0 & 0 \\ -3 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \overrightarrow{x_{n1}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \overrightarrow{x_{n2}} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda = 6 \rightarrow \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Find the eigenvalues and eigenvectors.

- **Then find a basis for the null space of** $(\lambda I - A)$**(the eigenspace) for each** λ**:**

$$\lambda = 0, A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -3 & 0 & -3 \\ 0 & -6 & 0 \\ -3 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 0 & -3 \\ 0 & -6 & 0 \\ -3 & 0 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} -3 & 0 & -3 \\ 0 & -6 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \rightarrow \vec{x}_{n1} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda = 0 \rightarrow \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Ex 2: Find the eigenvalues and eigenvectors.

- **Find the eigenvalues and a basis for each eigenspace for the matrix:**

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix}$$

Ex 2

- **First find the roots of $|\lambda I - A|$ (the eigenvalues):**

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 3 & 1 & -1 \\ -7 & \lambda + 5 & -1 \\ -6 & 6 & \lambda - 2 \end{vmatrix}$$

$$= (\lambda - 3)\begin{vmatrix} \lambda + 5 & -1 \\ 6 & \lambda - 2 \end{vmatrix} - 1\begin{vmatrix} -7 & -1 \\ -6 & \lambda - 2 \end{vmatrix} + (-1)\begin{vmatrix} -7 & \lambda + 5 \\ -6 & 6 \end{vmatrix}$$

$$= (\lambda - 3)\big((\lambda + 5)(\lambda - 2) - (-6)\big) - 1\big((-7)(\lambda - 2) - 6\big) - 1\big(-42 - (-6)(\lambda + 5)\big)$$

$$= (\lambda - 3)\big(\lambda^2 + 3\lambda - 10 + 6\big) - 1(-7\lambda + 14 - 6\ ) - 1(-42 + 6\lambda + 30\ )$$

$$= (\lambda - 3)\big(\lambda^2 + 3\lambda - 4\big) - 1(-7\lambda + 8) - 1(6\lambda - 12\ )$$

$$= \lambda^3 + 3\lambda^2 - 4\lambda - 3\lambda^2 - 9\lambda + 12 + 7\lambda - 8 - 6\lambda + 12$$

$$= \lambda^3 - 12\lambda + 16$$

Ex 2.

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow |\lambda I - A| = \lambda^3 - 12\lambda + 16$$

- **If there are integer solutions, they must be factors of the last term:** $\pm 2, \pm 4 \pm 8, \pm 16$

$$2^3 - 12(2) + 16 = 8 - 24 + 16 = 0 \rightarrow \lambda = 2 \rightarrow \lambda - 2 = 0$$

- **Keep testing or factor out the term you have found:**

$$\lambda^3 - 12\lambda + 16 = (\lambda - 2)(\lambda^2 + a\lambda + b) \rightarrow -2b = 16 \rightarrow b = -8$$

$$\lambda^3 - 12\lambda + 16 = (\lambda - 2)(\lambda^2 + a\lambda - 8) \rightarrow -8\lambda - 2a\lambda = -12\lambda \rightarrow a = 2$$

$$\lambda^3 - 12\lambda + 16 = (\lambda - 2)(\lambda^2 + 2\lambda - 8) = (\lambda - 2)(\lambda + 4)(\lambda - 2) = (\lambda - 2)^2(\lambda + 4)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 2)^2(\lambda + 4) = 0 \rightarrow \lambda = 2, -4$$

Ex 2

- **Then find a basis for the null space of** $(\lambda I - A)$ **(the eigenspace) for each** λ**:**

$$\lambda = -4, A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -7 & 1 & -1 \\ -7 & 1 & -1 \\ -6 & 6 & -6 \end{bmatrix}$$

$$\begin{bmatrix} -7 & 1 & -1 \\ -7 & 1 & -1 \\ -6 & 6 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} -7 & 1 & -1 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 \\ -7 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & -6 & 6 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \lambda = -4 \rightarrow \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Ex 2

- **Then find a basis for the null space of** $(\lambda I - A)$ **(the eigenspace) for each** λ**:**

$$\lambda = 2, A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -1 & 1 & -1 \\ -7 & 7 & -1 \\ -6 & 6 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & -1 \\ -7 & 7 & -1 \\ -6 & 6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 6 \\ 0 & 0 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \lambda = 2 \rightarrow \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Ex 3: Find the eigenvalues and eigenvectors.

- **Find the eigenvalues and a basis for each eigenspace for the matrix:**

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Ex 3

- **First find the roots of $|\lambda I - A|$ (the eigenvalues):**

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 1 & 1 & 1 & 1 \\ 0 & \lambda - 2 & 2 & 2 \\ 0 & 0 & \lambda - 3 & 3 \\ 0 & 0 & 0 & \lambda - 4 \end{vmatrix}$$

$$= (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4) = 0 \rightarrow \lambda = 1,2,3,4$$

- **Note: The eigenvalue of an upper (or lower) triangular matrix are simply the diagonal entries!**

Ex 3

- **Then find a basis for the null space of** $(\lambda I - A)$ **(the eigenspace) for each** λ**:**

$$\lambda = 1, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 0 & -1 & -1 & -1 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 & -1 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -1 & -1 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \lambda = 1 \rightarrow \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Ex 3

- **Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ:**

$$\lambda = 2, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \vec{x}_{n2} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \lambda = 2 \rightarrow \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Ex 3

- **Then find a basis for the null space of $\lambda I - A$ (the eigenspace) for each λ:**

$$\lambda = 3, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 2 & -1 & -1 & -1 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & -1 & -1 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -3/2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \vec{x}_{n1} = \begin{bmatrix} 3/2 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \lambda = 3 \rightarrow \left\{ \begin{bmatrix} 3 \\ 4 \\ 2 \\ 0 \end{bmatrix} \right\}$$

Ex 3

- **Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ:**

$$\lambda = 4, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 3 & -1 & -1 & -1 \\ 0 & 2 & -2 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1 & -1 & -1 \\ 0 & 2 & -2 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & -1 & 0 & -4 \\ 0 & 2 & 0 & -8 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & -1 & 0 & -4 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 0 & 0 & -8 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 0 & 0 & -8/3 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \vec{x}_{n1} = \begin{bmatrix} 8/3 \\ 4 \\ 3 \\ 1 \end{bmatrix}, \lambda = 4 \rightarrow \left\{ \begin{bmatrix} 8 \\ 12 \\ 9 \\ 3 \end{bmatrix} \right\}$$

Sec 6.2

Diagonalizing a matrix

Eigenvalues and eigenvectors

$$A\vec{x} = \lambda\vec{x}$$

An $n \times n$ matrix has at least 1 and at most n eigenvalues:

- **Solutions to $\det(\lambda I - A) = 0$ where λ is a variable.**
- **For each eigenvalue λ_i, the null space of $(\lambda_i I - A)$ is the eigenspace associated with λ_i; λ_i is a constant.**
- **The eigenspace is characterized by a basis; at least one vector for each distinct eigenvalue.**

Questions

Question 1: What is the relationship among the eigenvectors from different eigenvalues?

Answer: They are linear independent.

Question 2: How could we prove this?

Case 1: If there are only two vectors, prove that one vector is not a multiple of the other one.

Case 2: If there are more than two vectors, prove that one vector cannot be a linear combination of the other ones.

The eigenvectors of different eigenvalues are linearly independent

- Let $\vec{x}_1$, $\vec{x}_2$ and $\vec{x}_3$ be any three eigenvectors of A, and let λ_1, λ_2 and λ_3 be their corresponding eigenvalues:

$$A\vec{x}_1 = \lambda_1\vec{x}_1, \qquad A\vec{x}_2 = \lambda_2\vec{x}_2, \qquad A\vec{x}_3 = \lambda_3\vec{x}_3$$

- If $\vec{x}_2$ is a scalar multiple of $\vec{x}_1$ they must have the same eigenvalue:

$$\vec{x}_2 = k\vec{x}_1 \rightarrow A\vec{x}_2 = kA\vec{x}_1 = k\lambda_1\vec{x}_1 = \lambda_1\vec{x}_2 = \lambda_2\vec{x}_2 \rightarrow \lambda_1 = \lambda_2$$

- If $\vec{x}_1$, $\vec{x}_2$ and $\vec{x}_3$ are linearly dependent but not scalar multiples of each other, they must all have the same eigenvalue.

$$\vec{x}_3 = k\vec{x}_1 + l\vec{x}_2 \rightarrow A\vec{x}_3 = kA\vec{x}_1 + lA\vec{x}_2 = k\lambda_1\vec{x}_1 + l\lambda_2\vec{x}_2 = \lambda_3\vec{x}_3$$

$$k\lambda_1\vec{x}_1 + l\lambda_2\vec{x}_2 = k\lambda_3\vec{x}_1 + l\lambda_3\vec{x}_2 \rightarrow k(\lambda_1 - \lambda_3)\vec{x}_1 + l(\lambda_2 - \lambda_3)\vec{x}_2 = \vec{0}$$

$$\rightarrow \lambda_1 = \lambda_2 = \lambda_3$$

Case 1: n distinct eigenvalues.

If an $n \times n$ matrix has n distinct eigenvalues:

- **Each eigenvalue must generate exactly one linearly independent eigenvector.**
- **The set generated by taking one eigenvector for each eigenvalue is a basis for $\mathbb{R}^n$.**

Case 1: n distinct eigenvalues.

- **Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be distinct eigenvalues for an $n \times n$ matrix A, and let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ be any corresponding eigenvectors.**
- **We can express the eigenvalue equation in matrix form:**

$$A\vec{x}_i = \lambda\vec{x}_i. \quad \forall i \in [1, n] \rightarrow AX = X\Lambda$$

- **X is a matrix whose columns are the eigenvectors.**
- **Λ is a diagonal matrix whose entries are the corresponding eigenvalues.**

$$X = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n], \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix}$$

Case 1: n distinct eigenvalues.

$$\begin{aligned} AX &= A[\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n] \\ &= [A\vec{x}_1 \quad A\vec{x}_2 \quad \cdots \quad A\vec{x}_n] \\ &= [\lambda_1\vec{x}_1 \quad \lambda_2\vec{x}_2 \quad \cdots \quad \lambda_n\vec{x}_n] \end{aligned}$$

$$X\Lambda = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n]\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix} = [\lambda_1\vec{x}_1 \quad \lambda_2\vec{x}_2 \quad \cdots \quad \lambda_n\vec{x}_n]$$

$$A\vec{x}_i = \vec{x}_i, \forall i \in [1, n] \rightarrow AX = X\Lambda$$

Case 1: n distinct eigenvalues.

$$A\vec{x}_i = \vec{x}_i, \forall i \in [1, n] \rightarrow AX = X\Lambda$$

- X is a matrix whose columns are the eigenvectors; it will always be nonsingular as the columns must be linearly independent.
- Λ is a diagonal matrix whose entries are the corresponding eigenvalues; it will be nonsingular if and only if all the eigenvalues are nonzero.

$$AX = X\Lambda \rightarrow A = X\Lambda X^{-1}, \Lambda = X^{-1}AX$$

The Eigenvalue decomposition:

$$A = X\Lambda X^{-1}$$

Case 2: repeated eigenvalues.

In some cases an $n \times n$ matrix A, some of the eigenvalues are "repeated" when we factor $\det(\lambda I - A)$. The repeated eigenvalues may generate more than one linearly independent eigenvector or may not! (A may or may not be diagonalizable!!!!)

Definitions:

- **The geometric multiplicity of an eigenvalue is the dimension of the associated eigenspace (the number of linearly independent eigenvectors we can find).**
- **The algebraic multiplicity of an eigenvalue is number of times it is repeated when we factor the characteristic polynomial**

Examples of repeated eigenvalues

- **Example:** $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 \\ 1 & \lambda - 2 \end{vmatrix} = \lambda(\lambda - 2) - (-1) = (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

$$\lambda = 1 \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \rightarrow R = \vec{x}_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$ **is a basis for the eigenspace associated with** $\lambda = 1$**.**
- **There is a single eigenvalue with an algebraic multiplicity of 2 but a geometric multiplicity of 1.**

Examples of repeated eigenvalues.

- **Example:** $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 \\ 0 & \lambda - 3 \end{vmatrix} = (\lambda - 3)^2 = 0 \rightarrow \lambda = 3$$

$$\lambda = 3 \rightarrow (\lambda I - A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow R = \vec{x}_{n1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ **is a basis for the eigenspace associated with** $\lambda = 3$.
- **There is one eigenvalue with an algebraic and geometric multiplicity of 2.**
- **The eigenvectors span the entire** $\mathbb{R}^2$ **space.**

Diagonalization for repeated eigenvalues.

- **If an $n \times n$ matrix A, has n linearly independent eigenvectors, let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ be such a set, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be corresponding eigenvalues.**
- **We can express the eigenvalue equation in matrix form:**

$$A\vec{x}_i = \lambda\vec{x}_i \quad \forall i \in [1, n] \rightarrow AX = X\Lambda$$

- **X is a matrix whose columns are the eigenvectors.**
- **Λ is a diagonal matrix whose entries are the corresponding eigenvalues.**

$$X = [\vec{x}_1 \quad \vec{x}_2 \quad \cdots \quad \vec{x}_n], \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix}$$

- **Note 1: Without n independent eigenvectors, we cannot diagonalize.**
- **Note 2: Any matrix has no repeated eigenvalues can be diagonalized.**
- **Note 3: We can multiply eigenvectors by any nonzero scalars.**

Diagonalization: The Eigenvalue Decomposition (EVD)

Any $n \times n$ matrix A with n linearly independent eigenvectors can be expressed as:

$$A = X\Lambda X^{-1}$$

where X is a nonsingular matrix whose columns are the eigenvectors and Λ is a diagonal matrix whose entries are the corresponding eigenvalues.

Definition: A matrix that has a well-defined eigenvalue decomposition is said to be diagonalizable. Performing an eigenvalue decomposition is sometimes called diagonalizing the matrix.

Observation: A is diagonalizable if algebraic multiplicity= geometric multiplicity.

Diagonalization: Strang's Book

- Alternative Definition: (Strang's book)

Diagonalization Suppose the n by n matrix A has n linearly independent eigenvectors $x_1, \ldots, x_n$. Put them into the columns of an ***eigenvector matrix*** $\boldsymbol{X}$. Then $X^{-1}AX$ is the ***eigenvalue matrix*** $\boldsymbol{\Lambda}$:

Eigenvector matrix X
Eigenvalue matrix Λ

$$X^{-1}AX = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}. \tag{1}$$

Ex 1: The utility of the EVD: A^k

$$A = X\Lambda X^{-1} \rightarrow A^k = \prod_{i=1}^{k} A = AA\cdots A = (X\Lambda X^{-1})(X\Lambda X^{-1})\cdots(X\Lambda X^{-1})$$

$$A^k = X\left(\prod_{i=1}^{k}\Lambda\right)X^{-1} = X\Lambda^k X^{-1}$$

$$\Lambda^k = \left(\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix}\right)^k = \begin{bmatrix} (\lambda_1)^k & 0 & \cdots & 0 \\ 0 & (\lambda_2)^k & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & (\lambda_n)^k \end{bmatrix}$$

Ex 2: The utility of the $A\vec{x}_i = \vec{x}_{i+1}$

$$A = X\Lambda X^{-1}, A\vec{x}_i = \vec{x}_{i+1}$$

$$\rightarrow XA\vec{x}_i = X\vec{x}_{i+1} \rightarrow XAX^{-1}X\vec{x}_i = X\vec{x}_{i+1} \rightarrow \Lambda X\vec{x}_i = X\vec{x}_{i+1}$$

$$A\vec{x}_i = \vec{x}_{i+1} \rightarrow \Lambda\hat{x}_i = \hat{x}_{i+1}, \hat{x}_i = X\vec{x}_i$$

$$\Lambda\vec{u} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} \lambda_1 u_1 \\ \lambda_2 u_2 \\ \vdots \\ \lambda_n u_n \end{bmatrix}$$

Example: Matrix not diagonalizable

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

Their eigenvalues happen to be 0 and 0. Nothing is special about $\lambda = 0$, the problem is the repetition of λ. All eigenvectors of the first matrix are multiples of $(1, 1)$:

Only one line of eigenvectors $\quad A\boldsymbol{x} = 0\boldsymbol{x}$ means $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\boldsymbol{x} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

There is no second eigenvector, so this unusual matrix A cannot be diagonalized.

Similar Matrix

Definition: If B is an invertible matrix, then A and C are called similar matrix if it satisfies $A = BCB^{-1}$.

Theorem: All similar matrices have the same eigenvalues.

Proof Suppose $C\boldsymbol{x} = \lambda\boldsymbol{x}$. Then BCB^{-1} has the same eigenvalue λ with the new eigenvector $B\boldsymbol{x}$:

$$\textit{Same } \boldsymbol{\lambda} \qquad (BCB^{-1})(B\boldsymbol{x}) = BC\boldsymbol{x} = B\lambda\boldsymbol{x} = \lambda(B\boldsymbol{x}). \tag{5}$$

A fixed matrix C produces a family of similar matrices BCB^{-1}, allowing all B. When C is the identity matrix, the "family" is very small. The only member is $BIB^{-1} = I$. The identity matrix is the only diagonalizable matrix with all eigenvalues $\lambda = 1$.

Examples

Solving problems with the Eigenvalue Decomposition

Ex 1: Non-diagonalizable matrices.

- **Demonstrate why it is impossible to find a nonsingular X and diagonal Λ such that $A = X\Lambda X^{-1}$ where:**

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

Ex 1

- **Demonstrate why it is impossible to find a nonsingular X and diagonal Λ such that $A = X\Lambda X^{-1}$ where:**

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 2 & -3 & -4 \\ 0 & \lambda - 2 & -3 \\ 0 & 0 & \lambda - 2 \end{vmatrix}$$

$$= (\lambda - 2)(\lambda - 2)(\lambda - 2)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 2)^3 = 0 \rightarrow \lambda = 2$$

There is a single eigenvalue with an algebraic multiplicity of 3.

Ex 1

- **Demonstrate why it is impossible to find a nonsingular X and diagonal Λ such that $A = X\Lambda X^{-1}$ where:**

$$\lambda = 2, A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 0 & -3 & -4 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -3 & -4 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_n = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

- **The eigenspace of the only eigenvalue has dimension of 1 (which is less than 3); we can not find three linearly independent eigenvectors! So, we cannot complete the eigenvalue decomposition.**

Sec 6.4 and 6.5

Symmetric Matrices: The Spectral Decomposition and Positive Definite Matrices

The spectral theorem

Theorem: All symmetric (real) matrices: $A^T = A$:

- **Are always diagonalizable: all eigenvalues have a geometric multiplicity equal to the algebraic multiplicity.**
- **Always have eigenvalues that are exclusively real numbers.**
- **Always have eigenvectors corresponding to different eigenvalues that are orthogonal!**
  - **We can always construct an orthonormal set of eigenvectors!**

$$A = X\Lambda X^{-1} \rightarrow A = Q\Lambda Q^T,\ Q^T = Q^{-1}$$

Important: **We can always write this form, but we need to construct orthonormal bases for each eigenspace: normalization only if the dimension is 1, Gram-Schmidt otherwise!**

Orthogonal Eigenvectors Eigenvectors of a real symmetric matrix (when they correspond to different λ's) are always perpendicular.

Proof Suppose $S\boldsymbol{x} = \lambda_1 \boldsymbol{x}$ and $S\boldsymbol{y} = \lambda_2 \boldsymbol{y}$. We are assuming here that $\lambda_1 \neq \lambda_2$. Take dot products of the first equation with $\boldsymbol{y}$ and the second with $\boldsymbol{x}$:

$$\textbf{Use } S^{\mathrm{T}} = S \qquad (\lambda_1 \boldsymbol{x})^{\mathrm{T}} \boldsymbol{y} = (S\boldsymbol{x})^{\mathrm{T}} \boldsymbol{y} = \boldsymbol{x}^{\mathrm{T}} S^{\mathrm{T}} \boldsymbol{y} = \boldsymbol{x}^{\mathrm{T}} S \boldsymbol{y} = \boldsymbol{x}^{\mathrm{T}} \lambda_2 \boldsymbol{y}. \tag{3}$$

The left side is $\boldsymbol{x}^{\mathrm{T}} \lambda_1 \boldsymbol{y}$, the right side is $\boldsymbol{x}^{\mathrm{T}} \lambda_2 \boldsymbol{y}$. Since $\lambda_1 \neq \lambda_2$, this proves that $\boldsymbol{x}^{\mathrm{T}} \boldsymbol{y} = 0$. The eigenvector $\boldsymbol{x}$ (for λ_1) is perpendicular to the eigenvector $\boldsymbol{y}$ (for λ_2).

Complex eigenvalues and real matrices

For real matrices, complex λ's and x's come in "conjugate pairs."

$$\lambda = a + ib \qquad \bar{\lambda} = a - ib$$

$$\textit{If} \quad A\boldsymbol{x} = \lambda \boldsymbol{x} \quad \textit{then} \quad A\bar{\boldsymbol{x}} = \bar{\lambda}\bar{\boldsymbol{x}}. \tag{8}$$

Example 3 $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ has $\lambda_1 = \cos\theta + i\sin\theta$ and $\lambda_2 = \cos\theta - i\sin\theta$.

Those eigenvalues are conjugate to each other. They are λ and $\bar{\lambda}$. The eigenvectors must be $\boldsymbol{x}$ and $\bar{\boldsymbol{x}}$, because A is real:

$$\begin{aligned} &\textbf{This is } \lambda\boldsymbol{x} \qquad A\boldsymbol{x} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = (\cos\theta + i\sin\theta) \begin{bmatrix} 1 \\ -i \end{bmatrix} \\ &\textbf{This is } \bar{\lambda}\bar{\boldsymbol{x}} \qquad A\bar{\boldsymbol{x}} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = (\cos\theta - i\sin\theta) \begin{bmatrix} 1 \\ i \end{bmatrix}. \end{aligned} \tag{9}$$

Connection between elimination and eigenvalues

$$\textit{product of pivots} = \textit{determinant} = \textit{product of eigenvalues.}$$

However, the pivots may not the same as eigenvalues.

For symmetric matrices, the pivots and the eigenvalues have the same signs: The number of positive pivots = the number of positive eigenvalues.

Positive Definite Matrices

A symmetric matrix with ONLY POSITIVE eigenvalues is termed positive definite.

- **For a positive definite matrix:**

$$\vec{x}^T S \vec{x} > 0, \forall \vec{x} \neq \vec{0}$$

- **Note, if all eigenvalues are simply NON-NEGATIVE, the matrix is positive semi-definite, and we have**

$$\vec{x}^T S \vec{x} \geq 0, \forall \vec{x} \neq \vec{0}$$

Important application: the extremum test in calculus.

Application: the extremum test in Calculus.

Local extrema (maxima or minima) of a (differentiable) function of multiple variable are located where all partial derivatives are equal to zero:

$$\nabla F(\vec{x}^*) = \left[\frac{\partial F(\vec{x}^*)}{\partial x_i}\right] = \vec{0}$$

The Hessian matrix is a matrix of all the second partial derivatives:

$$H(\vec{x}^*) = \left[\frac{\partial^2 F(\vec{x}^*)}{\partial x_i \partial x_j}\right]$$

The Hessian is symmetric everywhere the second derivatives are continuous.

- **The Hessian has a well-defined spectral decomposition:** $H(\vec{x}^*) = Q\Lambda Q^T$
- **If all eigenvalues are positive, the Hessian is positive definite:**
  - **The extremum is a local minimum.**
- **If all eigenvalues are negative, the Hessian is *negative definite*:**
  - **The extremum is a local maximum.**

Ex 1: Find the spectral decomposition.

- **If possible, find orthogonal Q and diagonal Λ such that where $A = Q\Lambda Q^T$.**

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Find the spectral decomposition.

- **If possible, find orthogonal Q and diagonal Λ such that where $A = Q\Lambda Q^T$.**

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 & -1 \\ 0 & \lambda - 4 & 0 \\ -1 & 0 & \lambda - 3 \end{vmatrix}$$

$$= -0 + (\lambda - 4)\begin{vmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{vmatrix} - 0 = (\lambda - 4)\left((\lambda - 3)^2 - 1\right)$$

$$= (\lambda - 4)\left(\lambda^2 - 6\lambda + 9 - 1\right) = (\lambda - 4)\left(\lambda^2 - 6\lambda + 8\right) = (\lambda - 4)(\lambda - 4)(\lambda - 2)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 4)^2(\lambda - 2) = 0 \rightarrow \lambda = 2,4$$

- **There one eigenvalue with an algebraic multiplicity of 2 and one with a multiplicity of 1.**

Find the spectral decomposition.

- **If possible, find orthogonal Q and diagonal Λ such that where $A = Q\Lambda Q^T$.**

$$\lambda = 2, A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ -1 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_n = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

- **We want an orthonormal basis:**

$$\vec{q}_{\lambda=2} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Find the spectral decomposition.

- **If possible, find orthogonal Q and diagonal Λ such that where $A = Q\Lambda Q^T$.**

$$\lambda = 4, A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

- **We want an orthonormal basis.**

$$(\vec{x}_{n2})^T \vec{x}_{n1} = 0 \rightarrow \vec{q}_{\lambda=4,1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{q}_{\lambda=4,2} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Find the spectral decomposition.

- **If possible, find orthogonal Q and diagonal Λ such that where $A = Q\Lambda Q^T$.**

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \rightarrow \lambda = 2,4, \vec{q}_{\lambda=4,1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{q}_{\lambda=4,2} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \vec{q}_{\lambda=2} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Ex 2: Using the eigenvalue decomposition (EVD).

- Given the spectral decomposition:

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & {}^{1}/_{\sqrt{2}} & {}^{-1}/_{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & {}^{1}/_{\sqrt{2}} & {}^{1}/_{\sqrt{2}} \end{bmatrix}$$

- Find a general expression for $C = A^k$.
- Find a matrix such that $B^2 = A$.

Using the eigenvalue decomposition.

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

- **Find a general expression for $C = A^k$.**

$$C = A^k = Q\Lambda^k Q^T = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 4^k & 0 & 0 \\ 0 & 4^k & 0 \\ 0 & 0 & 2^k \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 4^k & 0 \\ 4^k/\sqrt{2} & 0 & 4^k/\sqrt{2} \\ -2^k/\sqrt{2} & 0 & 2^k/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 4^k/2 + 2^k/2 & 0 & 4^k/2 - 2^k/2 \\ 0 & 4^k & 0 \\ 4^k/2 - 2^k/2 & 0 & 4^k/2 + 2^k/2 \end{bmatrix}$$

Using the eigenvalue decomposition.

$$A = Q\Lambda Q^T, \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

- **Find a matrix such that $B^2 = A$. Let $B = Q\Lambda^{1/2}Q^T$ then $B^2 = Q\Lambda Q^T = A$**

$$B = Q\Lambda^{1/2}Q^T = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 \\ \sqrt{2} & 0 & \sqrt{2} \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 + 1/\sqrt{2} & 0 & 1 - 1/\sqrt{2} \\ 0 & 2 & 0 \\ 1 - 1/\sqrt{2} & 0 & 1 + 1/\sqrt{2} \end{bmatrix}$$